# Project 1 – ME5250 Robotics Mechanics and Control

# Retracting Track Mechanism

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**Abstract**

Retracting wheel tracks on mobile robots presents a significant kinematic challenge due to the inherent friction generated when tracks are moved perpendicular to their primary direction of motion. This project presents a novel multi-linkage retraction mechanism designed by me that addresses this challenge through coordinated deployment of auxiliary support wheels during the retraction sequence. The mechanism consists of two primary sub-systems: an 10-bar linkage optimized for compact track retraction, and a 4-bar linkage that deploys auxiliary wheels perpendicular to the track direction during retraction to minimize dragging friction. The two system coordinates to deploy wheel at full extension. A functional 3D-printed prototype successfully demonstrates the mechanism's ability to achieve smooth track retraction while mitigating friction through strategically timed auxiliary wheel engagement. Singularity analysis identifies stable operating regions and confirms the absence of dead-point configurations within the intended range of motion.

**Introduction**

Mobile robots with continuous track systems often require the ability to retract tracks toward the body for navigation through confined spaces or storage. However, track retraction presents a fundamental challenge: while tracks move tangentially during normal operation, retraction requires perpendicular motion, generating substantial dragging friction that can exceed actuator limits, cause excessive wear, and prevent successful retraction.

This project presents a self-designed multi-linkage retraction mechanism inspired by Chebyshev Lambda linkage principles. The design employs two coordinated sub-mechanisms: a compact 10-bar linkage for track retraction and a 4-bar linkage that deploys auxiliary support wheels exclusively during retraction. These auxiliary wheels, oriented perpendicular to the track direction, temporarily lift the tracks to significantly reduce dragging friction. The 4-bar mechanism automatically retracts the auxiliary wheels when tracks reach full extension, ensuring they don't interfere with normal operation. Comprehensive mobility and kinematic analyses are performed on both sub-mechanisms, complemented by a detailed SolidWorks CAD model and a functional 3D-printed prototype demonstrating the full range of motion and system operation.

**Literature Survey**

Started with exploring various track mechanisms. The study "Mechanism Design and Motion Ability Analysis for Wheel/Track Mobile Robot" examines how a hybrid wheel-track mobile robot uses a linkage to adjust its running gear geometry, enabling reconfiguration between wheel and track modes. In parallel, the paper "Design and Analysis of a Planar Six-Bar Crank-Driven Leg Mechanism for Walking Robots" presents a six-bar linkage that generates a foot path with distinct phases (support, lift, return) from a single crank input demonstrating

the viability of six-bar mechanisms for sequential motion cycles. "Graphical synthesis of 6-bar dwell linkage mechanism" shows how six-bar mechanisms (Stephenson topology) can be designed so the output link dwells (i.e., remains stationary) for a portion of the input cycle. The idea was to have a cyclic motion be converted to a unique output motion which can be utilised at the output.

Later explored multiple straight bar linkage mechanisms. Examples include the Chebyshev λ-linkage, a four-bar mechanism developed by Pafnuty Chebyshev that provides a long nearly-straight coupler motion segment, and the Hoecken linkage which provides an extended straight motion with very nearly constant velocity through its linear portion. These mechanisms underscore how a rotary input can be converted into a controlled, sustained linear (or near-linear) motion. Also explored an 10-bar mechanism by Jeremy Fielding used in a similar robot system to retract tracks. This design was adapted as it provided compact packaging with reasonable retraction reach. Along with this Chebyshev λ-linkage was used in conjunction to retract and deploy the smaller drag reducing wheel.

**Mechanism Design Overview**

During track extension, the primary retraction linkage reaches its fully extended configuration, causing the Chebyshev-inspired mechanism's coupler point to reach its maximum horizontal displacement and begin its return stroke. This downward motion of the Chebyshev mechanism's long coupler arm drives a compound joint assembly consisting of a slot-follower mechanism mounted on a revolute joint, which itself is housed within a prismatic guide. As the coupler arm descends, the geometric constraint imposed by the slot forces rotation about the revolute joint, generating the retraction motion of the auxiliary wheel assembly. This mechanical coupling ensures that auxiliary wheel deployment occurs exclusively during track retraction, with automatic retraction occurring as the tracks return to their extended position.

Key Components:
10-bar retraction linkage: Its purpose is to act as an actuator for the tract to move in and out while also remaining compact. This mechanism although has too many degrees of freedom to be used alone, provides support to the overall structure of the robot.

4-bar auxiliary wheel deployment linkage: Provides unique end effector workspace to achieve desired outcome.

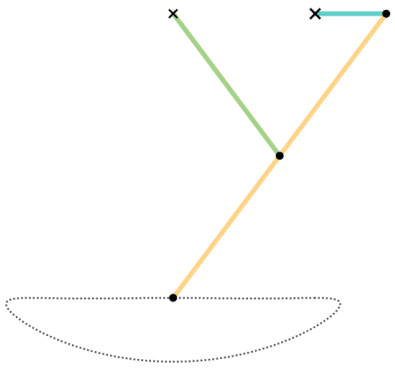


*Figure 1 Chebyshev Lambda Linkage*

Connecting system: This is a minor system which acts as the link between the above mentioned two linkages. This system includes a prismatic joint, a revolute joint anda  slot joint.

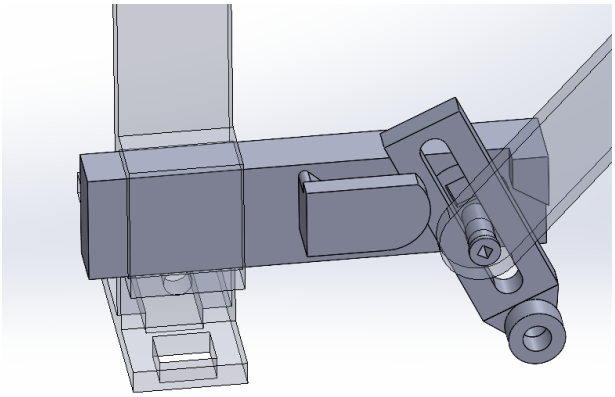


*Figure 2 Connecting system*

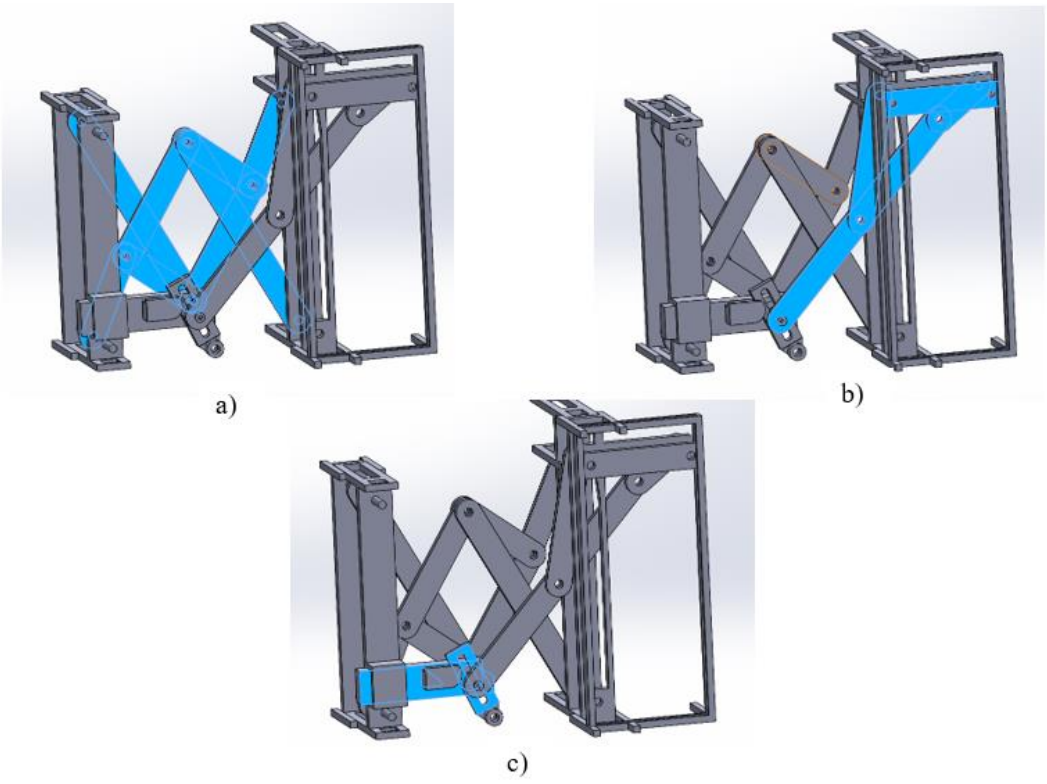



*Figure 3 a) 10-bar linkage, b) Chebyshev 4-bar linkage, c) prismatic and slot joint*

**Analysis:**

***Grubler's Analysis:***

10-bar linkage: Considering it as a planar mechanism with 10 links and 12 revolute joints each of degree 1.

$$DOF = m(N - 1 - J) + \sum_{1}^{J} f_i$$

$$DOF = 3(10 - 1 - 12) + 12 = \ 3$$

The system has 3 DOF but considering the robot to be on the ground one more prismatic joint is added between the ground base link and the final link. Therefore,

$$DOF = 3(10 - 1 - 13) + 13 = \ 1$$

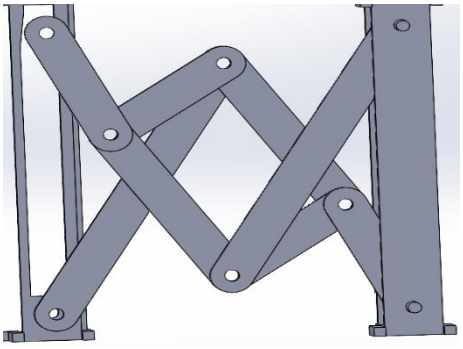


*Figure 4: 10-bar linkage with 10 links and 12 joints*

The current 10 bar system analysis indicates that the system requires contact at both ends to achieve a satisfactory result. The only advantage this system provides is compactness when folded.

Chebyshev 4 bar linkage: This is a 4 link 4 joint system with all revolute joints.

$$DOF = 3(4 - 1 - 4) + 4 = 1$$

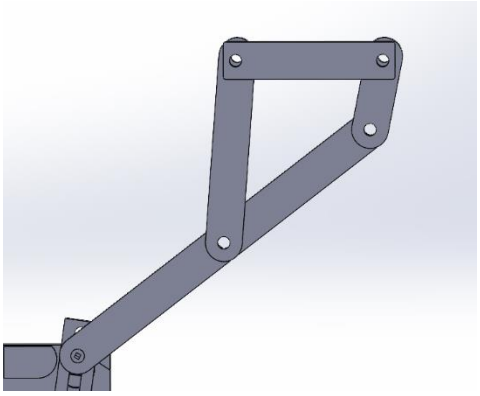


*Figure 5 Chebyshev mechanism*

As the equation suggests, the system has 1 DOF.

Overall system:

The system is supposed to be considered as a planar mechanism but due to design constraints it was made as a spatial mechanism. We will do analysis for Planar Grubler.

It has 15 links and 19 joints, 18 are of 1 degree and 1 joint is of 2 degrees. Therefore f=20

$$DOF\ (planar\ floating) = 3(15 - 1 - 19) + 20 = 5$$

For grounded mechanism we add an extra prismatic joint to simulate moving along the ground

$$DOF = 3(15 - 1 - 20) + 21 = 3$$

One of the degrees is in the direction of retraction. Another one is for the rotation of the connecting mechanism, and the last one is for the prismatic movement at the end of the connecting mechanism.

**Forward Kinematics**

*Chebyshev:*

Position of point B:

$$\mathrm{B} = (b\cos\ \theta_2, b\sin\ \theta_2)$$

Geometric constraint:

$$\| \mathrm{B} + d \cdot [\cos\ \theta_4, \sin\ \theta_4] - [a, 0] \| = c$$

Law of cosines at point B:

$$\cos\ (\angle B) = \frac{d^2 + \| \mathrm{B} - \mathrm{O}_3 \|^2 - c^2}{2d \cdot \| \mathrm{B} - \mathrm{O}_3 \|}$$

Rocker angle:

$$\theta_4 = \psi \pm \arccos\ (\cos\ (\angle B))$$

where

$$\psi = \arctan\ 2(-y_B, a - x_B)$$

Position of point C (midpoint):

$$\mathrm{C} = \mathrm{B} + d \cdot [\cos\ \theta_4, \sin\ \theta_4]$$

Position of point E (auxiliary wheel):

$$\mathrm{E} = \mathrm{B} + d_{full} \cdot [\cos\ \theta_4, \sin\ \theta_4]$$

**Range of motion and workspace analysis**:

Kinematic analysis reveals that the input crank can oscillate through approximately ~ 200° of angular displacement, bounded by dead-point configurations at $\theta_2$_min and $\theta_2$_max. The endpoint (point E) traces an arc-like trajectory with a vertical displacement of 30 mm and horizontal displacement of 140 mm. This range of motion is specifically designed to allow the auxiliary wheel to deploy downward during track retraction, lifting the tracks slightly off the ground to reduce perpendicular dragging friction, and then retract when tracks reach full extension.

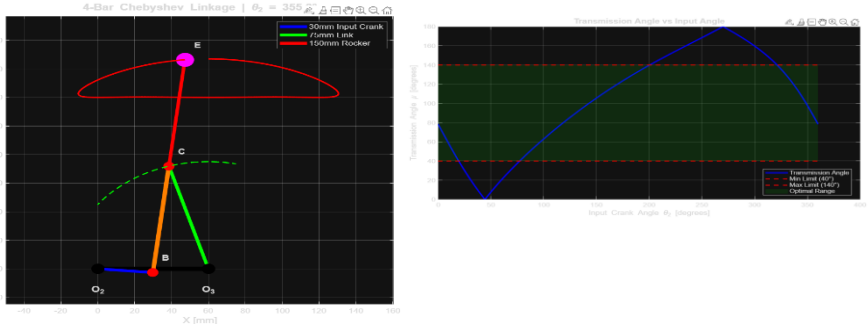



*Figure 6: Graph of Range of motion*

**Singularities**:

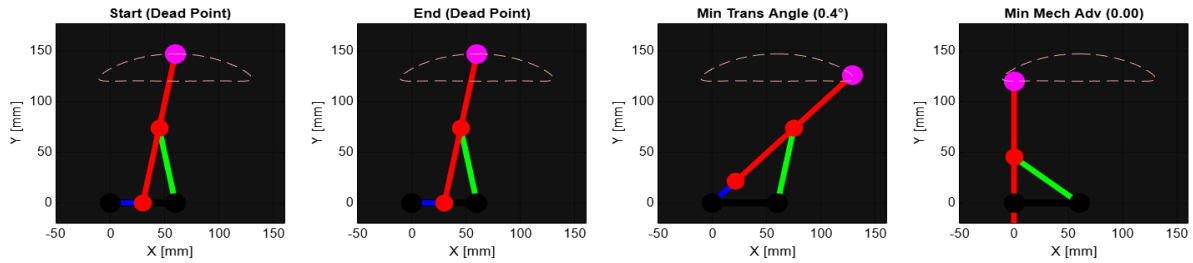



*Figure 7: Graphs of Singularity positions*

**Prototype:**

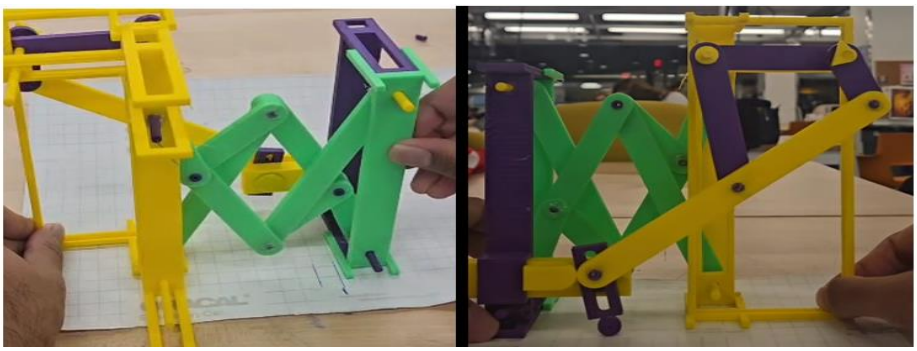


*Figure 8: 3D printed model*

**Result and Conclusion**

Mobility Analysis: Gruebler-Kutzbach analysis (n=4, j=4) confirmed DOF = 1, requiring single actuator input. Grashof criterion (S+L = 180mm > P+Q = 135mm) identified the mechanism as a non-Grashof double-rocker with limited oscillation, appropriate for controlled deployment.

Forward Kinematics: Loop-closure equations successfully mapped input angle $\theta_2$ to all joint positions. The auxiliary wheel (point E) demonstrated adequate vertical displacement for ground contact during track retraction and proper clearance when retracted. Trajectory analysis confirmed correct deployment timing through purely mechanical coordination.

Singularity Analysis: Dead-point singularities at configuration boundaries provide natural mechanical stops. Transmission angles remained within acceptable ranges ($40° < \mu < 140°$) throughout most of the workspace. Jacobian analysis showed no internal singularities, confirming smooth velocity transmission.

Prototype Validation: 3D-printed prototype successfully demonstrated full range of motion, coordinated auxiliary wheel deployment, and validated the friction-reduction concept. Physical testing confirmed analytical predictions.

**Conclusion**:
This project developed a combined system of a 4-bar linkage mechanism and 10 bar linkage mechanism that addresses track retraction friction by deploying auxiliary support wheels during perpendicular motion.

Achieved sufficient range of motion for effective operation, favourable singularity characteristics with dead points at natural boundaries, and validated prototype performance. The mechanism enables reduced actuator requirements and decreased component wear.

Limitations include unfavourable DOF under certain scenarios, requirement of proper timing belts for coordination between the two mechanisms. Unwanted torque when actuated by single actuation point.

The project validates a functional proof-of-concept design that offers a potential solution for the track retraction challenge, which could become a reliable solution for mobile robotics applications in the future.

Future development aims to integrate both sub-mechanisms into a unified 1-DOF retraction system and transition from planar to spatial linkage configurations to reduce the overall footprint currently occupied by the Chebyshev mechanism.

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